Status Externalities in Education and Low Birth Rates in Korea

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American Economic Review, 114(6): 1576–1611 (2024)

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ECON Paper Presentation

Low fertility and high education spending

- Korea's total fertility rate was **0.78 in 2022**. The Korean government has expressed concerns over low birth rates, as the latter imply **rapid population aging** and pose a considerable challenge to the **public pension system**.
- Education spending per child accounts for **9.2% of family income**.
  Education Fever: also occurs in other East Asian countries, such as China.
  Status Externality: Parents value the education of their children relative to the education of other children.

Question: Can competition over children's education help explain low fertility?

Appendix: Sample and fertility measures
Appendix: Education spending measures

Education is a heavier burden for poorer families

Income quintile	1st	2nd	3rd	4th	5th
Children per family	1.80	1.91	1.87	1.93	2.03
Childless (%)	5.3	4.0	2.0	1.3	2.0
Education / income (%)	11.2	9.9	9.3	8.7	6.9

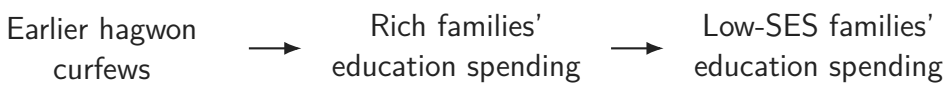
Education spending is per child; quintiles use long-term family income.

- Lower-income families have fewer children and higher childlessness.
- They spend a larger **share of income** on each child's education. These associations alone do not establish causality.

Appendix: Fertility and childlessness by income

How does one family's spending affect another's?

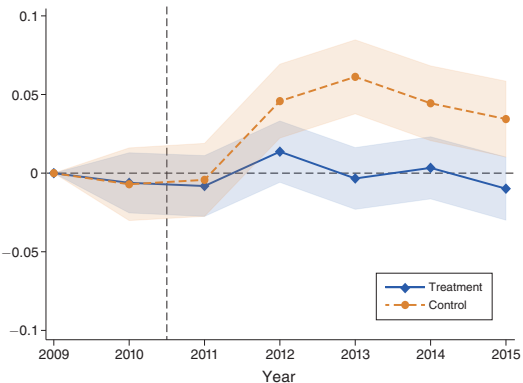
We need empirical evidence:
Results: $\beta = 0.039$: a 10 percent reduction in the top 15 percent income households' spending on private education decreases spending for the lower half of the education distribution by around 0.39 percentage points.
Methods:



- Use curfew differences across provinces and years, 2009–2015.
- Low-SES children attend late-night classes less often.

Appendix: IV equations and identification

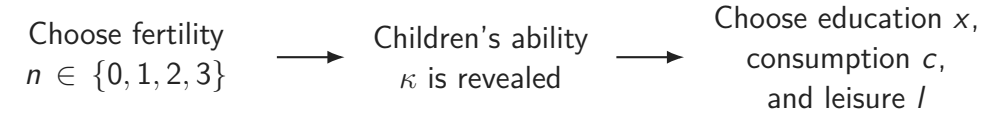
Evidence of education-spending spillovers



- Pre-policy trends are not distinguishable in the plot.
- Normalized log spending; 95% confidence bands. The IV does not estimate fertility effects.

Appendix: IV estimates across specifications

Parents choose fertility and education together



- Families differ in parents' human capital, inherited ability, and preferences for consumption relative to children.
- Each child requires education spending, childcare time, and additional household consumption.
- Parents value children's human capital, including their position relative to other children.

Appendix: Household optimization problem

Competition raises the cost of children

Parents value children's human capital relative to a benchmark:

$$\phi(n)\log\left(h' - \chi\widetilde{h}'\right).$$

- h' is a child's human capital; $\widetilde{h}'$ is average human capital among children of the richest 15% of families.
- A higher benchmark raises the incentive to invest per child. With limited resources, parents may choose fewer children.
- Education also produces skills: $h' = A_h\kappa(\theta + x^\alpha)$. Reducing education spending can therefore harm future productivity.

Appendix: Equilibrium feedback and output

Calibrating fertility, education, and income

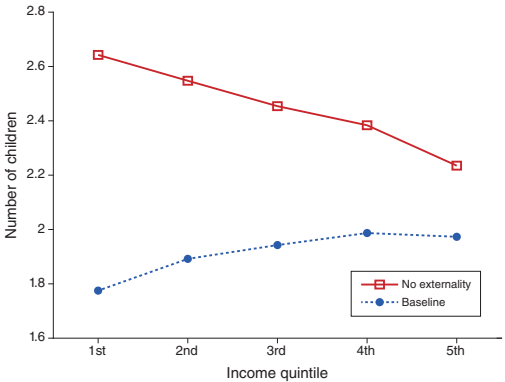
- Eleven parameters jointly target twelve moments, including fertility, education spending, inequality, and intergenerational mobility.
- The calibrated status parameter is $\chi = 0.078$. The spending spillover and low-income childlessness help discipline it.

Selected target moments	Model	Data
Education-spending spillover	.038	.039
Childless in bottom income quintile	.052	.053
Mean education spending / income	.110	.092

Fit is imperfect. These are targeted moments, not independent validation.

Appendix: Calibration targets and parameter values
Appendix: Calibration objective and external parameters

Removing status concerns raises fertility



- Setting $\chi = 0$ raises average fertility from 1.92 to 2.45, about 28%.
- Bottom-quintile childlessness falls from 5.2% to 0.2%.
- The fertility–income gradient becomes negative.

This is a model counterfactual. It does not measure the share of Korea's historical fertility decline explained by status competition.

Appendix: Removing status concerns versus freezing the benchmark

Lower childcare costs have a smaller effect

Model experiment	Fertility	Change
Baseline	1.92	—
Reduce childcare time cost λ by 4.3%	1.97	+2.9%
Remove status concerns ($\chi = 0$)	2.45	+28%

- The childcare-cost experiment approximates the effect of changing social norms through a lower cost of children.
- The model does not explicitly represent the division of childcare between mothers and fathers. This is a limited comparison of mechanisms.

Appendix: Childcare-cost counterfactual
Appendix: Fertility across birth cohorts

Two policies target different household choices

Instrument	Household incentive	Financing
Child allowance	Reduces the net cost of an additional child	Uniform lump-sum tax
Education tax	Raises the price of private education	Revenue returned as a uniform rebate

- The allowance acts on fertility directly; the education tax changes the cost of investing in each child.
- Both policies also change other families' spending and the equilibrium benchmark.

Appendix: Government budgets and household constraints

Higher fertility, lower investment and output

	Baseline	Child allowance		Education tax	
		$\psi = .01$	$\psi = .02$	$\tau_x = .10$	$\tau_x = .20$
Fertility	1.92	2.00	2.08	1.97	2.01
Change (%)	—	+4.2	+8.5	+2.9	+4.8
Education / income (%)	11.0	10.4	9.9	8.9	7.4
Mean human capital	3.29	3.24	3.20	3.18	3.11
Output per capita	.958	.938	.918	.901	.860

New steady states. Education is per child; human capital and output use model units.

- Both policies raise fertility. The education tax reduces human capital and output more in these experiments.

Appendix: Why the income elasticities change

The objective is current parents' welfare

- The planner maximizes weighted welfare of the **initial generation**, using Negishi weights across households.
- The available instruments are a child allowance and a tax on education. Government budgets balance in every period.
- The reform is permanent, and households anticipate the transition. The initial generation keeps its existing human capital.
- Future generations' utilities are not separately included in the objective. Their welfare must therefore be examined after finding the optimum.

Appendix: Welfare weights and optimization
Appendix: Interpreting welfare gains and efficiency

Poorer families respond more to the optimal mix

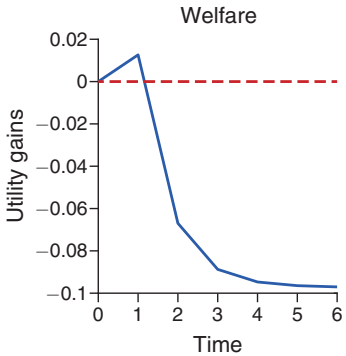
- Optimal education tax: $\tau_x = 21.6\%$; child allowance: $\psi = .0175$ (about US$61 per month on the paper's 2015 scale).
- Initial-generation fertility rises by 10.8%, while education investment per child falls by 39.3%.

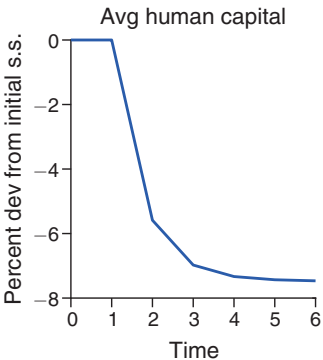
Income quintile	1st	2nd	3rd	4th	5th	All
Fertility change (%)	23.2	15.6	9.1	5.4	2.8	10.8
Investment change (%)	−45.5	−42.9	−40.3	−37.7	−35.6	−39.3

Effects refer to the initial generation. Investment is per child among families with children; percentage changes are the authors' reported values.

Appendix: Converting the allowance into monthly dollars

Later generations lose on average





Reform begins in period 1 (25 years per period). Welfare: weighted utility changes.

- Parents gain consumption and leisure; their children inherit less human capital.

Conclusion

- Education competition raises the cost of children, with stronger fertility effects among poorer families in the model.
- Child allowances and education taxes can raise fertility while lowering education investment and future human capital.
- A policy that benefits current parents can leave future generations worse off. The challenge is to reduce competition while preserving education's productive benefits.

Appendix: IV evidence
Appendix: Household model
Appendix: Calibration
Appendix: Welfare objective

A1. Sample and fertility measures

- The completed-fertility sample contains **756 KLIPS households**: married or cohabiting couples with women born in 1970–1975.
- Households have at least three observations while the woman is aged 40–43. Completed fertility counts children ever born.
- Long-term income averages household labor and capital income over these ages, excluding social insurance and transfers.
- The 2022 total fertility rate of 0.78 is a period statistic. It is different from the sample's completed fertility of 1.91.

Return to: Low fertility and high education spending

A2. Education spending measures

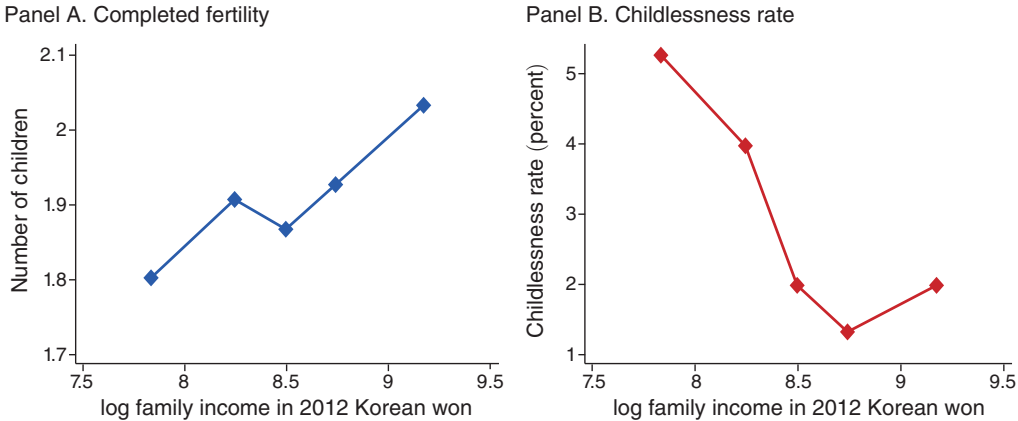
- The individual survey measures private education by child and school stage. The household survey includes broader out-of-pocket spending, including public-school tuition.
- The quantitative exercise rescales the income gradient to the comprehensive lifetime mean of 9.2%.

Income quintile	1st	2nd	3rd	4th	5th
Private education / income (%)	8.4	7.4	7.0	6.5	5.1
Rescaled measure (%)	11.2	9.9	9.3	8.7	6.9

These ratios are per child. The IV outcome instead divides education spending by total household expenditure, not income.

Return to: Low fertility and high education spending

A3. Fertility and childlessness by income



Two-adult households grouped by long-term income; descriptive differences.

Return to: The income gradient

A4. IV specification and identification

$$\log E_{st} = \delta_{10}\Lambda_{st}^{10} + \delta_{11}\Lambda_{st}^{11} + \gamma' X_{ist} + \epsilon_{ist},$$
$$\vartheta_{ist} = \beta \log \widehat{E}_{st} + \eta' X_{ist} + \varepsilon_{ist}.$$

- E_{st}: top-income families' per-student spending by province and year (PEES).
- ϑ_{ist}: education per student / household expenditure (KLIPS).
- Curfews: 10 pm and 11 pm, relative to midnight. Controls: income, parental education and age, province effects, and time controls.
- Exclusion requires no direct curfew effect on low-SES spending.

A 10% rich-spending decline implies $100 \times 0.039 \log(0.9) \approx -0.41$ percentage points.

Return to: Curfew identification Appendix: IV estimates

A5. IV estimates across six specifications

	Low income			Low education		
	(1)	(2)	(3)	(4)	(5)	(6)
Spillover β	.065	.091	.049	.039	.070	.042
Standard error	(.014)	(.030)	(.038)	(.015)	(.039)	(.043)
Linear time trend	No	Yes	No	No	Yes	No
Year fixed effects	No	No	Yes	No	No	Yes
Observations		1,501			1,528	

Column (4), in teal, supplies the calibration target. Standard errors use the wild clustered bootstrap. Year fixed effects reduce precision.

Return to: Spending-spillover evidence

Appendix: First-stage estimates Appendix: Spillover mapping

A6. First-stage results and instrument strength

	Low income			Low education		
	(1)	(2)	(3)	(4)	(5)	(6)
10 pm curfew	−.146	−.110	−.092	−.148	−.106	−.089
Standard error	(.026)	(.029)	(.031)	(.024)	(.025)	(.028)
11 pm curfew	−.060	−.031	−.033	−.054	−.019	−.019
Standard error	(.039)	(.037)	(.035)	(.035)	(.033)	(.031)
F-statistic	16.6	7.5	4.4	20.0	9.2	5.2
Linear time trend	No	Yes	No	No	Yes	No
Year fixed effects	No	No	Yes	No	No	Yes

Year fixed effects weaken the instruments. Reference: midnight. Standard errors in parentheses.

Return to: Spending-spillover evidence

A7. The household optimization problem

Choose fertility before ability is revealed:

$$\max_{n\in\{0,1,2,3\}} \mathbb{E}_{\kappa|\kappa^p} V(h, b, \kappa, n; \widetilde{h}).$$

Then choose consumption, education, and leisure:

$$V = \max_{c,x,l} \left\{ b \log \frac{c}{\Lambda(n)} + \nu \frac{l^{1-\gamma}}{1-\gamma} + \phi(n) \log\left(h' - \chi \widetilde{h}'\right) \right\},$$

$$c + nx \leq wh(1 - \lambda n - l), \quad h' = A_h \kappa(\theta + x^{\alpha}),$$

$$0 \leq l \leq 1 - \lambda n, \qquad \widetilde{h}' = \Gamma(\widetilde{h}).$$

$c > 0$, $x \geq 0$; require $h' > \chi \widetilde{h}'$ for $n > 0$. The child term is zero for $n = 0$. Siblings share the same ability.

Parents value children's human capital (warm glow). $\Lambda(n)$ scales consumption.

Return to: Household choices

A8. Equilibrium feedback and aggregate output

- Parents' education choices determine the next generation's human capital and the benchmark faced by other households.
- Equilibrium expectations must match this implied benchmark. Fertility weights the next generation's population distribution.

$$L_t = \mathbb{E}[h(1 - \lambda n_t - l_t)], \qquad Y_t = AL_t, \qquad A = 1.$$

- L_t is labor in efficiency units; the expectation averages over the current population and ability realizations.
- A steady state reproduces the same **normalized distribution** and benchmark. Population size can still change.

Return to: Education competition mechanism

A9. Calibration: fertility and inequality

Parameter	Value	Target moment	Model	Data
ϕ_1	1.39	Share with 1 child	.176	.196
ϕ_2	2.18	Share with 2 children	.633	.631
ϕ_3	2.57	Share with 3+ children	.158	.144
σ_κ	.361	Income Gini	.266	.263
ν	1.87	Average hours worked	.290	.301
σ_b	.587	Income elasticity of fertility	.080	.082

Parameters jointly minimize relative gaps across twelve moments. Row pairings indicate identification intuition, not separate estimation.

Return to: Calibration overview

Appendix: Status concerns and human-capital parameters

A10. Calibration: status and human capital

Parameter	Value	Target moment	Model	Data
χ	.078	Education investment spillover	.038	.039
		Bottom-quintile childlessness	.052	.053
θ	1.08	Mean investment / income	.110	.092
α	.668	Income elasticity of investment	.775	.698
ρ_κ	.376	Intergenerational income elasticity	.327	.320
A_h	2.23	Output per capita	.958	1.00

The model overpredicts education spending. Output is approximately normalized: .958 is the calibrated model level, not exactly 1.

Return to: Calibration overview

Appendix: Calibration objective and external parameters

A11. Calibration method and external parameters

$$\widehat{p} = \arg\min_{p} \sum_{j=1}^{12} \left[\frac{m_j^{\text{model}}(p) - m_j^{\text{data}}}{m_j^{\text{data}}} \right]^2 .$$

- Eleven parameters jointly fit twelve moments.
- External choices include $\gamma = 2$, childcare time cost $\lambda = .041$, and production productivity $A = 1$.
- Use the OECD household scale, at most three children, and twenty grid points each for preferences and ability.
- Targeted moments assess fit, not out-of-sample performance.

Return to: Calibration overview

A12. Mapping the spending spillover

Perturb top-income families' spending by $\Delta_{top} = -0.1$:

$$\bar{h}'_{top,0} = A_h \bar{\kappa}_{top}(\theta + \bar{x}^{\alpha}_{top}),$$
$$\bar{h}'_{top,1} = A_h \bar{\kappa}_{top}\left[\theta + \left((1+\Delta_{top})\bar{x}_{top}\right)^{\alpha}\right].$$

Scale the benchmark by $\widetilde{h}_1 = (\bar{h}'_{top,1}/\bar{h}'_{top,0})\widetilde{h}_0$, re-solve choices, and compute the printed response expression:

$$100\frac{\mathbb{E}[\widetilde{\vartheta}_1 \mid \Upsilon_{bottom}] - \mathbb{E}[\widetilde{\vartheta}_0 \mid \Upsilon_{bottom}]}{\Delta_{top}}, \qquad \widetilde{\vartheta}_i = \frac{x_i}{c_i + x_i n_i}.$$

Υ_{bottom}: below-median income. The factor 100 converts share changes into percentage points; regression and response units must agree.
Response moment: $\beta^{model} = .038$. Preference parameter: $\chi = .078$.

Return to: Spending-spillover evidence

A13. From model transfers to monthly dollars

- Inputs: 2015 annual GDP per capita of US$28,732, a 25-year model period, and payments over 18 years of childhood.
- Scale the transfer by calibrated model output, $Y = .958$:

$$M = \frac{\psi}{.958}\frac{28{,}732 \times 25}{12 \times 18}.$$

Allowance	Monthly US$ per child	Monthly GDP per capita (%)
$\psi = .0100$	34.71	1.45
$\psi = .0175$	60.75	2.54

This uses the paper's 2015 income scale. It is not a current-dollar policy quotation.

Return to: Optimal policy mix

A14. Government budgets

Child allowance only:

$$c + xn + T \leq wh(1 - \lambda n - l) + \psi n, \qquad T = \psi \mathbb{E}[n].$$

Education tax only:

$$c + (1 + \tau_x)xn \leq wh(1 - \lambda n - l) + R, \qquad R = \tau_x \mathbb{E}[xn].$$

Combined policy, with T defined as a lump-sum tax:

$$T = \psi \mathbb{E}[n] - \tau_x \mathbb{E}[xn].$$

- Lump-sum payments can affect choices through income effects.
- Expectations average over households and ability. Here R denotes the rebate; the paper uses T for both roles.

Return to: Policy instruments

A15. Status concerns and benchmark feedback

Setting $\chi = 0$ removes status concerns. Fixing $\widetilde{h} = 5.78$ retains the motive but removes its feedback.

	Baseline	Transfer $\psi = .02$		Tax $\tau_x = .20$	
Benchmark feedback	Yes	Yes	No	Yes	No
Fertility	1.92	2.08	2.06	2.01	1.97
Fertility change (%)	—	8.5	7.4	4.8	3.0
Childlessness (%)	3.4	2.9	3.1	3.3	3.5
Education / income (%)	11.0	9.9	10.1	7.4	7.7
Average labor supply	.290	.286	.287	.275	.277
Average human capital	3.29	3.20	3.21	3.11	3.13

Equilibrium feedback amplifies the fertility responses in these experiments.

Return to: Removing status concerns

A16. The childcare-cost counterfactual

All households	Baseline	Lower childcare cost
Completed fertility	1.92	1.97
Childlessness (%)	3.4	3.2
Education per child / income (%)	11.0	10.7

- Reducing childcare time cost λ by 4.3% raises fertility by a reported 2.9%.
- This approximates a change in norms through a single cost parameter. It does not explicitly model the allocation of childcare within the couple.
- The result is specific to this experiment and calibration.

Return to: Childcare costs and status concerns

A17. Fertility across birth cohorts

Birth cohort / quintile	1st	2nd	3rd	4th	5th
Children per family					
1961–1966	1.99	1.97	2.06	2.13	2.08
1970–1975	1.80	1.91	1.87	1.93	2.03
Childlessness (%)					
1961–1966	3.17	.79	.79	.00	.79
1970–1975	5.26	3.97	1.99	1.32	1.99

- The highest-income quintile has the smallest fertility decline.
- This descriptive comparison among couples is not a causal test of norms. Household income is not mother's education.

Return to: Childcare costs and status concerns

A18. Welfare weights and the policy objective

$$\max_{\tau_x,\psi} \mathbb{E}_{F_1}\left[\omega(h, b, \kappa^p, \kappa)\, V_1(h, b, \kappa, n_1; \widetilde{h}_1)\right].$$

- Maximize initial-generation welfare subject to equilibrium behavior, consistent expectations, and balanced budgets.
- The permanent reform keeps τ_x and ψ constant from period 1.
- To construct Negishi weights, estimate on baseline simulated data:

$$\log c = \beta_0 + \beta_1 \log h + \beta_2 \log \kappa^p + \beta_3 \log b + \beta_4 \log \kappa + \epsilon.$$

$$\omega \propto \frac{\widehat{c}}{b}, \qquad \text{normalized to sum to one.}$$

Predicted consumption $\widehat{c}$ is used to approximate inverse marginal utility.

Return to: The welfare criterion

A19. Interpreting welfare gains and efficiency

- The reported gain is **0.0126 weighted utility units**. The authors' reference-agent conversion is 1.2% of lifetime consumption.
- This is not a uniform gain for every household. Figure 5 compares average weighted welfare across generations.
- The optimum is restricted to two policy instruments and values the initial generation; it is not an unrestricted optimum for all generations.
- The theoretical $\mathcal{A}$-efficiency result compares people alive in both allocations and requires the relevant uniqueness and positive-weight conditions.

Return to: Welfare across generations
Return to: The welfare criterion

A20. The college-admission probability

Replace the child-utility term with $\phi(n)\Pi(h';\widetilde{h}')$, where

$$\Pi(h';\widetilde{h}') = \max\left\{0, 1 - e^{-\iota_2(h'+\iota_0-\iota_1\widetilde{h}')}\right\}.$$

- The zero-probability threshold is $h^* = \iota_1\widetilde{h}' - \iota_0$. Above it, admission probability rises toward one.
- ι_1 controls the strength of relative competition; ι_0 shifts the threshold; ι_2 controls the rise above it.
- The function approximates competition for college admission. It does not impose an exactly fixed quota of places.

Return to: The tournament interpretation
Appendix: Tournament policy experiments

A21. Policies in the tournament model

Steady-state experiment	Baseline	$\psi = .02$	$\tau_x = .20$
Fertility	1.90	2.08	1.88
Fertility change (%)	—	+9.5	−1.0
Education / income (%)	9.6	9.0	7.9

- Optimal mix: $\tau_x = 36.3\%$, $\psi = .0244$. Initial-generation fertility rises 11.9%; investment per child falls 38.1%.
- College-going children per family increase from 1.45 to 1.50, a reported 3.7% increase. This is roughly stable, not an exactly constant seat count.

Return to: The tournament interpretation

A22. Interpreting the income elasticities

- The elasticity describes how an average outcome differs with 1% higher income.
- The income gradients need not be causal; fertility is discrete.

New steady states	Baseline	$\psi = .01$	$\psi = .02$
Income elasticity of fertility	.080	.020	$-.021$
Income elasticity of investment	.775	.824	.868

- Poorer families increase fertility more and reduce investment per child more, flattening the first gradient and steepening the second.
- The IV coefficient β is a spending-share response, not an income elasticity.

Return to: Steady-state policy effects